Modeling Transactions at Corporación Favoritas Grocery Stores Over Time

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Abstract

The Corporación Favoritas grocery stores in Ecuador have kept data on the total number of daily transactions at their stores from 2013 to 2017. To choose hours of operation at these stores, it may be worthwhile to model transactions over time while taking into account the effect of weekends and holidays on store performance. In this report, we fit time series models to two stores located in Quito and Guayaquil. We found that holidays and weekends had differing effects on transactions at the two stores, and that the mean number of transactions at the store in Guayaquil was increasing over time. However, both stores showed strong positive correlation and seasonal cycling.

Introduction

For many companies and retail stores, scheduling hours in order to maximize profit while minimizing costs is the most important consideration for many thriving businesses. The Corporación Favoritas stores in Ecuador are no exception. Over the span of 4 years from 2013 to 2017, the number of daily transactions at each Favoritas store was recorded. Our analysis can be broken down into a few primary focuses. First, we modeled transactions over time at two different stores — Store 1 in Quito and Store 28 in Guayaquil — using regression. We fit seasonal models to the data that took into account the effects of weekends and holidays on transactions. After fitting the best model for both stores, we compared the effect that weekends and holidays had on transactions using our regression model. Finally, we used spectral analysis to identify important features and limitations of our models. We will start this report by explaining the technical details involved in this analysis. Next, we will give a brief overview of the numerical results of the analysis. Finally, we will interpret these results in the context of the research questions that were originally posed along with discussing possible limitations and future directions of this work.

Data

Our data consists of the total number of daily transactions at 54 Corporación Favoritas grocery stores in Ecuador, recorded by the stores from 2013 to 2017. The dataset includes the date (our predictor variable) and the total number or products purchased in the store on that day (our response variable).

Each of the 54 stores are located in 1 of 22 cities in Ecuador. For our analysis, we have chosen to analyze Store 1, which is located in Quito, and Store 28, which is located in Guayaquil. We will be looking at the total number of products sold daily, as well as the total number of products sold by week at these stores.

Our dataset, 'transactions.csv', was found on kaggle.com under the title "Store Sales - Time Series Forecasting".

Results

We first analyzed the results for store 1 in Quito. First observing the time series plot, as shown in figure 1, we can see that there appears to be a weekly cycle of store sales being lower during the weekends as opposed to week days for store 1. Additionally, there is a spike every year that appears to be during the week before Christmas in late December.

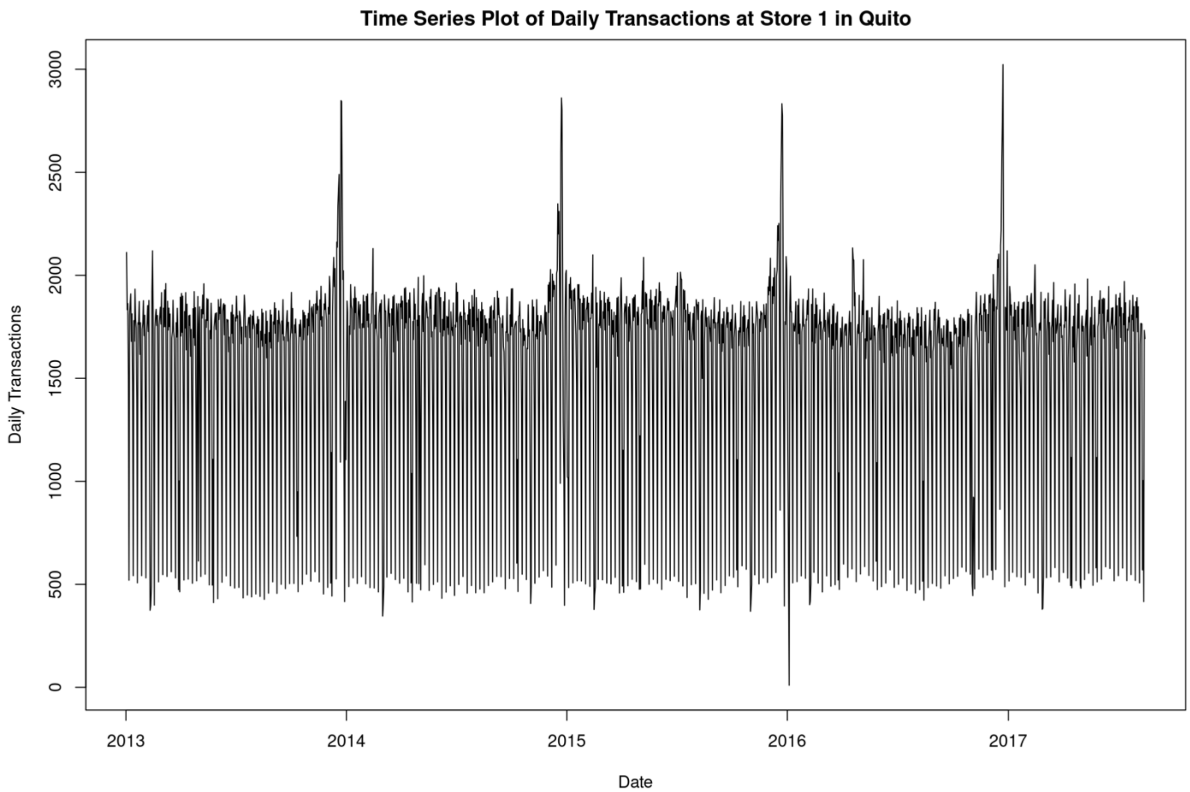

Figure 1. Time series plot of daily transactions at Store 1 in Quito, Ecuador.

There appeared to be a drop from weekday sales to sales on Saturdays, followed by an even larger drop to Sunday sales. After temporarily removing all points from Saturdays and Sundays and observing these new points, there appears to be additional days that have drops in sales in a semi-seasonal manner, as seen in figure 2.

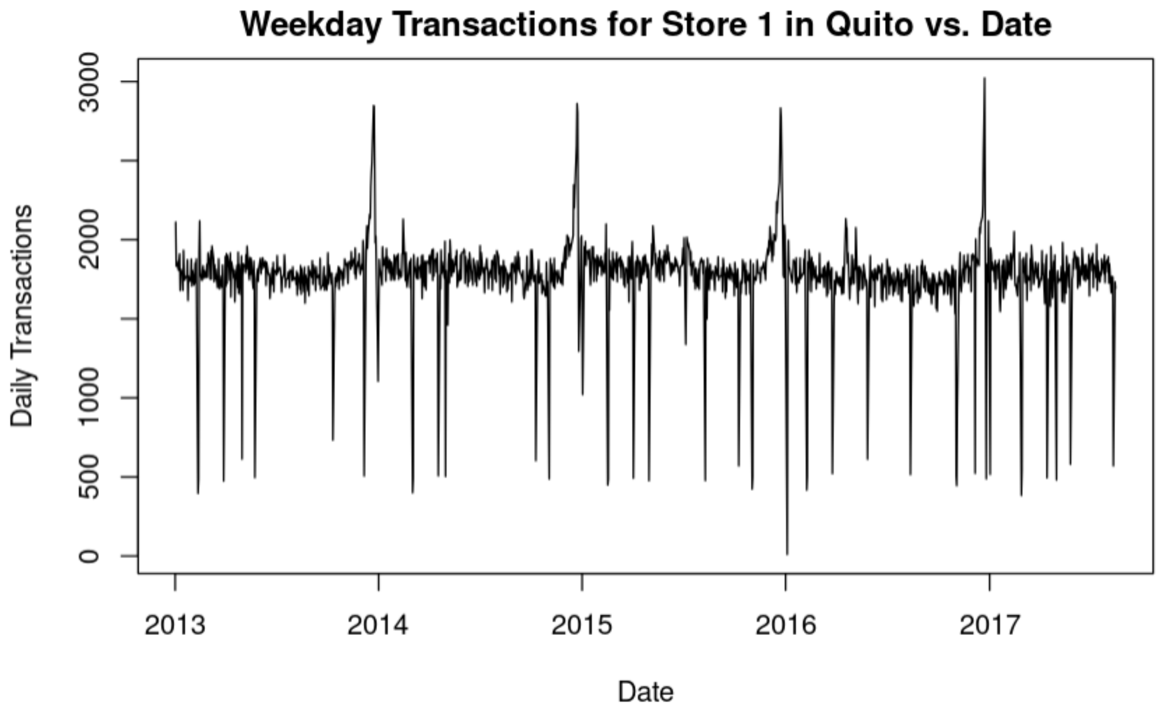

Figure 2. Weekday transactions for store 1 that shows dips mainly during holidays

Of the days that have less than 1000 transactions, we found that almost all of these days corresponded to an Ecuadorian holiday. Excluding New Year's and Christmas since these days are not in the dataset (likely due to the store being closed on these days), the following holidays were found to be correlated with less sales (the exact dates of these holidays change every year):

- Carnival, February 11-12, 2013
- Good Friday, March 29, 2013
- Labor Day, May 1, 2013
- Battle of Pichincha Day, May 24, 2013
- First Cry of Independence, August 10, 2013
- Guayaquil Independence Day, October 11, 2013
- Day of the Dead, November 1, 2013

- All Souls Day, November 2, 2013
- Cuenca Independence Day, November 4, 2013

With this, we created the variables for our mean regression model. The variables included the date (in decimal form), if the day was Saturday, if the day was Sunday, if it was a holiday, and if it was the week before Christmas (December 18-24).

We then aimed to find the SARIMA model that best fit our mean regression model. Based on the plot of the residuals, and the ACF and PACF for the residuals, we found that an ARMA(1,1) x $(1,1)_7$ best fit our data. We figured that this model was the best fit because the ACF and PACF of the residuals both showed exponential decay, with a seasonal term every 7 lags that also displayed exponential decay in the ACF and PACF of the residuals. Additionally, this model had the lowest AIC value of the models we tested. While the AIC was still high and the residual ACF and PACF still showed some possible significant correlation, this was still one of the best models we were able to find. The plots for these are in the code appendix below.

	ϕ_1	θ_1	$\mathbf{\Phi_1}$	$\mathbf{\Theta_1}$	isSaturday	isSunday	isHoliday	isChristmasWeek	decimal_date
ARMA(1, 1) x $(1, 1)_7$	0.3216	-0.1725	0.9607	-0.8985	-485.7253	-1278.8588	-1000.0953	674.6223	-5.8579
S.E.	0.1506	0.1571	0.0169	0.0275	16.6375	16.5238	22.3497	31.7919	6.9914

Table 1. Regression model estimates and coefficients with ARMA(1, 1) x $(1, 1)_7$ errors for Store 1 transactions.

From our mean regression model as shown in table 1, we can see that if the day was Sunday is associated with the largest drop in transactions, with Sundays being associated with a mean drop of 1273 transactions compared to other days. Holidays were also associated with a large drop in transactions, with holidays being associated with a mean drop of 1002 transactions.

We then plotted the periodogram versus the fitted ARMA spectrum we found to be the best fit. From figure 3, we can see that the model captures our data fairly well, especially the weekly peaks, but the fitted spectrum overestimates the variability of the lowest frequencies, so the fitted spectrum overestimates the long-term meandering from the mean for store 1.

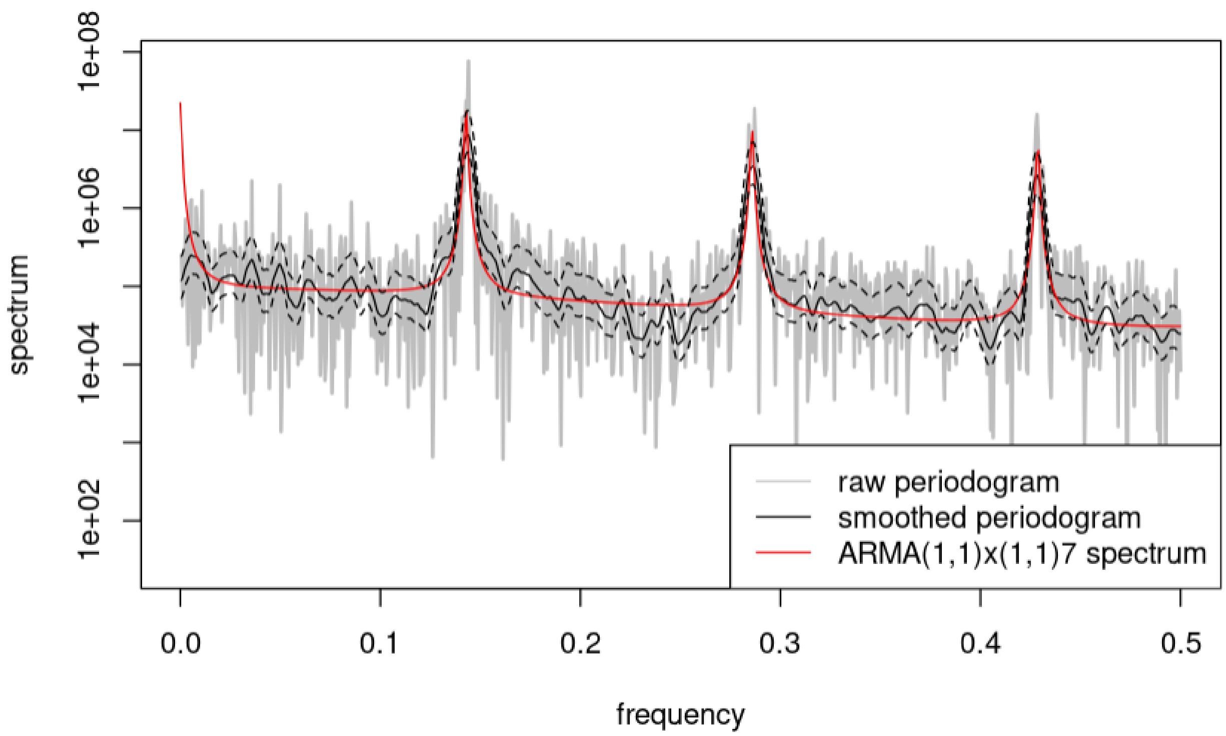

Figure 3. The periodogram versus the fitted ARMA(1,1)x$(1,1)_7$ spectrum for store 1

We then analyzed the results for store 28 in Guayaquil. First observing the time series plot, as shown in figure 4, we can see that there appears to be a weekly cycle of store sales being higher during the weekends as opposed to week days for store 1. Additionally, there is a spike every year that appears to be during the week before Christmas in late December, with Christmas Eve and New Year's Eve especially showing prominent spikes.

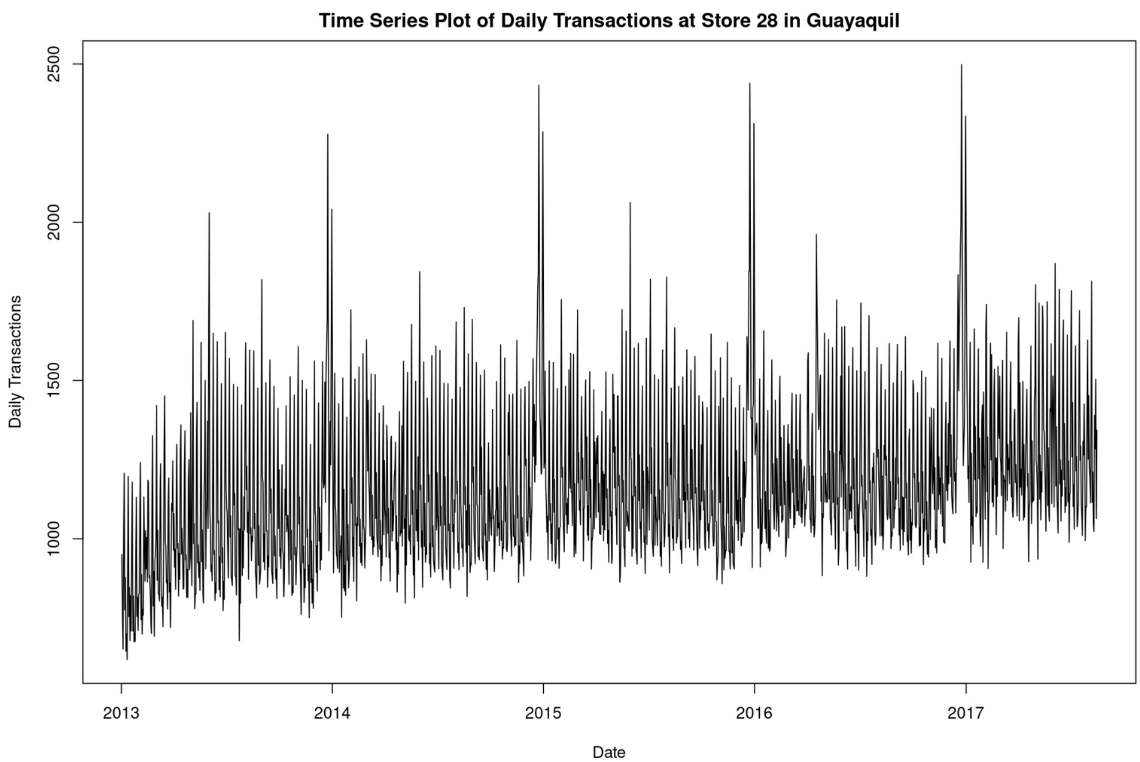

Figure 4. Time series plot of daily transactions at Store 28 in Guayaquil, Ecuador.

With this, we created the variables for our mean regression model. The variables included the date (in decimal form), if the day was Saturday, if the day was Sunday, if it was a holiday, if it was the week before Christmas (December 18-23), if it was Christmas Eve, and if it was New Years Eve. We used the same holidays as the ones used in store 1 for consistency.

We then aimed to find the SARIMA model that best fit our mean regression model. Based on the plot of the residuals, and the ACF and PACF for the residuals, we found that an ARIMA(2,0,1) x $(2,0,1)_7$ best fit our data. We figured that this model was the best fit because the ACF and PACF of the residuals both showed exponential decay with the ACF more clearly being significant after 2 lags, with a seasonal term every 7 lags that also displayed exponential decay in the ACF and PACF of the residuals. Additionally, this model had the lowest AIC value of the models we tested.

	ϕ_1	ϕ_2	θ_1	$\mathbf{\Phi_1}$	$\mathbf{\Phi_2}$	$\mathbf{\Theta_1}$	isSaturday	isSunday	isHoliday	isChristmasWeek
ARMA(2, 1) x $(2, 1)_7$	0.0444	0.2094	0.2941	0.8549	0.0930	-0.7354	215.6350	374.6297	215.6350	394.7489
S.E.	0.2086	0.0755	0.2131	0.0467	0.0368	0.0410	18.1309	19.3375	15.4373	31.4414

	isDec24	isDec31	decimal_date
ARMA(2, 1) x $(2, 1)_7$	1151.4442	875.9442	67.7601
S.E.	51.0294	50.6615	14.2310

Table 2. Regression model estimates and coefficients with ARMA(2, 1) x $(2, 1)_7$ errors for Store 28 transactions.

From our mean regression model as shown in table 2, we can see that if the day was December 24 is associated with the largest boost in transactions, with December 24th being associated with a mean boost of 1151 transactions compared to other days. Sundays were also associated with a decently large boost in transactions, with Sunday being associated with a mean boost of 364 transactions.

We then plotted the periodogram versus the fitted ARMA spectrum we found to be the best fit. From figure 5, we can see that the model captures our data alright, but the fitted spectrum underestimates the variability of the frequencies across most of the spectrum. The fitted spectrum has a similar structure of the peaks and weekday trends of the periodogram, but consistently underestimates variability.

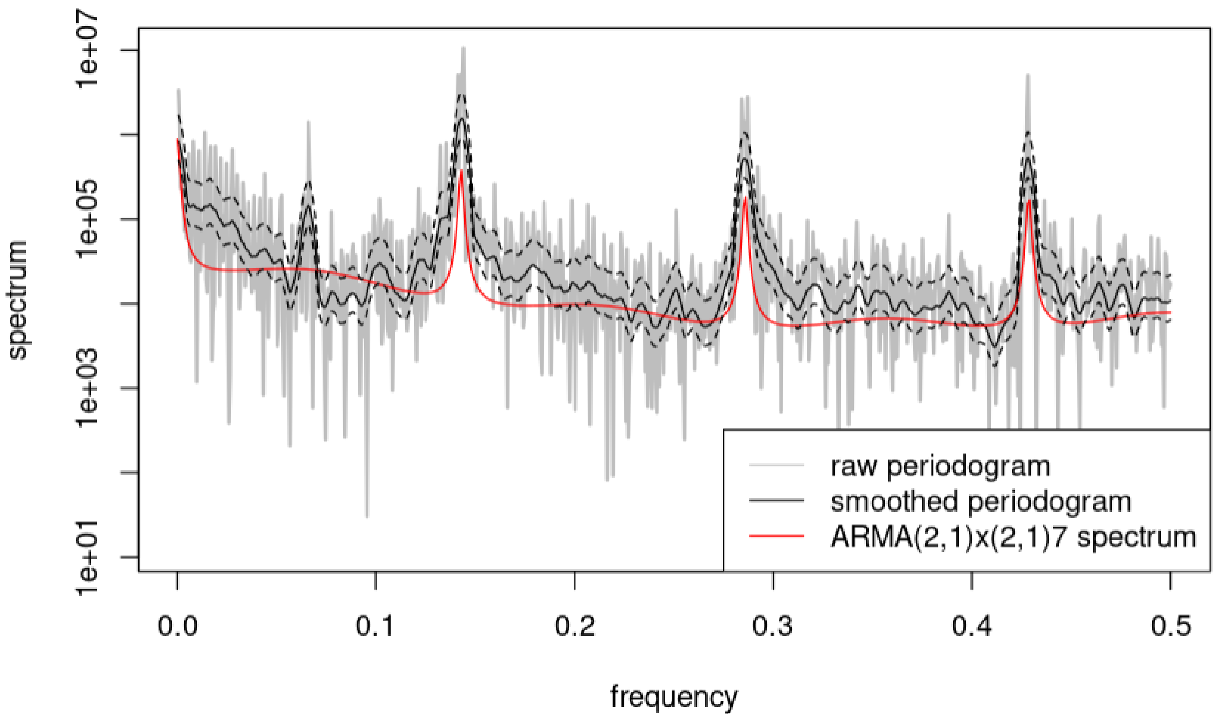

Figure 5. The periodogram versus the fitted ARMA(2,1)x$(2,1)_7$ spectrum for store 28

Discussion

We observed several differences in the best models for Store 1 and Store 28. For one, Store 28 showed an increasing number of transactions over time while Store 1 showed transactions remaining fairly constant. Additionally, weekends and holidays had opposite effects on transactions for Store 1 and Store 28. For Store 1, transactions decreased on the weekends and on holidays. For Store 28, transactions increased on the weekends and on holidays. These discrepancies may have been a result of different weekend and holiday hours between the two stores. Store 28 may have had longer hours on weekends and holidays, while Store 1 may have had reduced hours during these days. To choose the hours of operations at these stores, our models indicate that the lesser sales on holidays and weekends for store 1 and longer hours of operation may be less beneficial, while the greater sales on holidays and weekends indicate that longer hours of operation for store 28 may be more beneficial.

There were also several similarities between the best models for the two stores. For both stores, transactions increased during the week before Christmas. Additionally, both stores showed a strong seasonal cycle, with a repeating pattern every 7 days. Finally, we saw evidence of strong positive temporal correlation for both stores. Thus, we saw less variation in transactions day-to-day. However, the same model can not be used to model transactions between these two stores, but both stores could benefit from increased hours during the week leading up to Christmas.

A limitation of our analysis is that both fitted ARMA models were not very sufficient models. The ACF and PACF of these models indicated that some lags still had statistically significant correlations, so there was room to be desired for the sufficiency of both models. It's possible that better fitted models exist, and we were not able to find them. Another limitation is that more variables could have helped make our mean regression model more accurate, so adding more variables such as specific variables for each holiday could possibly contribute to more accurate models. Future research could be conducted to address these limitations, or could analyze the other stores in Ecuador to come to a more broad conclusion.

References

Kaggle. Store Sales - Time Series Forecasting.

https://www.kaggle.com/competitions/store-sales-time-series-forecasting/overview